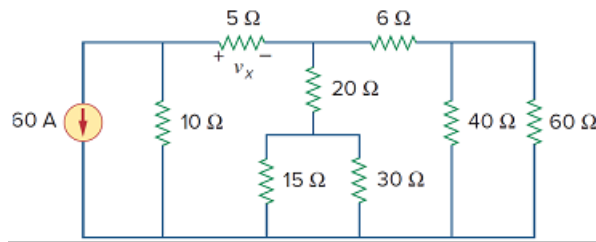


Problem

In the circuit shown in Fig. 2.87, determine V_x and the power absorbed by the $60\text{-}\Omega$ resistor.

Figure 2.87:



Step-by-step solution

Step 1 of 6

Refer to Figure 2.87 in the textbook.

In the circuit, the resistor $40\text{ }\Omega$ is in parallel with $60\text{ }\Omega$ and its equivalent is in series with $6\text{ }\Omega$. Find its equivalent resistance.

$$\begin{aligned} R_{eq1} &= 6 + [40 \parallel 60] \\ &= 6 + \left[\frac{(40)(60)}{40 + 60} \right] \\ &= 6 + 24 \\ &= 30\text{ }\Omega \end{aligned}$$

Similarly, the resistor $15\text{ }\Omega$ is in parallel with $30\text{ }\Omega$ and its equivalent is in series with $20\text{ }\Omega$. Find its equivalent resistance.

$$\begin{aligned} R_{eq2} &= 20 + [15 \parallel 30] \\ &= 20 + \left[\frac{(15)(30)}{15 + 30} \right] \\ &= 20 + 10 \\ &= 30\text{ }\Omega \end{aligned}$$

Step 2 of 6

Draw the reduced circuit diagram.

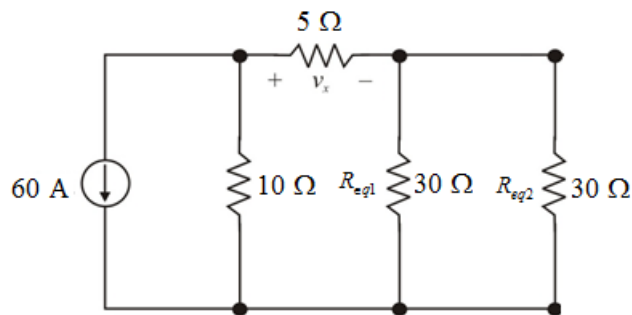


Figure 1

Step 3 of 6

In Figure 1, the resistor R_{eq1} is in parallel with R_{eq2} . Its equivalent resistance is,

$$\begin{aligned} R_{eq3} &= 30 \parallel 30 \\ &= \frac{(30)(30)}{30+30} \\ &= 15 \Omega \end{aligned}$$

Draw the modified circuit diagram.

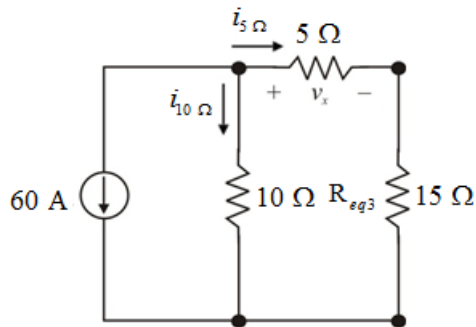


Figure 2

Step 4 of 6

Use Current division rule to find the current through 5Ω resistor.

$$\begin{aligned} i_{5\Omega} &= \left(\frac{10}{10 + (5 + 15)} \right) (-60) \\ &= \left(\frac{1}{3} \right) (-60) \\ &= -20 \text{ A} \end{aligned}$$

Use Ohm's law to find the voltage across 5Ω resistor.

$$v_x = 5i_{5\Omega}$$

Substitute -20 A for $i_{5\Omega}$.

$$\begin{aligned} v_x &= (5)(-20) \\ &= -100 \text{ V} \end{aligned}$$

Therefore, the value of unknown voltage, v_x in the circuit is -100 V .

Step 5 of 6

Refer to Figure 1. The current $i_{5\Omega}$ is equally divided between the two 30Ω resistors. That is

$\frac{i_{5\Omega}}{2}$ current is flowing through each 30Ω resistor. This $\frac{i_{5\Omega}}{2}$ current is again divided between the parallel resistors 40Ω and 60Ω . Find the current flowing through 60Ω resistor.

$$\begin{aligned} i_{60\Omega} &= \left(\frac{40}{40 + 60} \right) \left(\frac{i_{5\Omega}}{2} \right) \\ &= \left(\frac{40}{100} \right) \left(\frac{-20}{2} \right) \\ &= -4 \text{ A} \end{aligned}$$

Step 6 of 6

Determine the power absorbed by the $60\ \Omega$ resistor.

$$\begin{aligned} p_{60\ \Omega} &= (i_{60\ \Omega}^2)(60) \\ &= (-4)^2(60) \\ &= 960\ \text{W} \end{aligned}$$

Therefore, the power absorbed by the $60\ \Omega$ resistor is 960 W.